Shengao Yi

Department of City and Regional Planning, University of Pennsylvania, PA 19104, US

Email: shengao@upenn.edu | Tel: +1 2673241376 Website Google Scholar LinkedIn

EDUCATION

University of Pennsylvania — Philadelphia, US
Doctor of Philosophy in City and Regional Planning — Aug 2024 – May 2027 (Expected)
GPA: **4.0**/4.0
Advisor: Prof. Xiaojiang Li

University of Pennsylvania — Philadelphia, US
Master of Science in Urban Spatial Analytics — Aug 2022 – May 2023
GPA: **3.81**/4.0

Shenzhen University — Shenzhen, China
Bachelor of Engineering in Geospatial Information Engineering — Aug 2018 – Jun 2022
GPA: **85**/100
Advisor: Prof. Wei Tu

RESEARCH INTERESTS

- **AI for Environmental/Landscape Planning, Optimization and Design**
- **Micro-scale Urban Analytics**
- **Geospatial Artificial Intelligence (GeoAI)**
- **Climate Modeling**

AWARDS

- Penn AI Fellowship ($8000 annually) — Jan 2026
- Penn IUR Undergraduate Urban Research Colloquium ($2000) — Jan 2026
- Third Prize of 2022 Smart City Research and Innovation Scheme (SCRIS) — July 2023
- 2022 Excellent Graduation Thesis (Shenzhen University) — Jun 2023
- The Third Prize of 2021 Super Map Cup University GIS Competition — Oct 2021
- Software Copyright: Electric Unmanned Networked Fleet Intelligent Operation and Monitoring Prototype System. — Dec 2022
- Software Copyright: Parade Car and Online Car-hailing Operation Analysis System — Nov 2022
- Shenzhen University "Star of Innovation and Entrepreneurship" — Nov 2022
- The Fourth Winner of Big Data Track Creativity Award in 2020 Digital China Innovation Competition. — Dec 2021

REFEREED JOURNAL PUBLICATIONS (*Corresponding Author)

1. **Yi, S.,** Li, X., Wang, L., Tu, W., Cui, Q., Zhang, Y., ... & Zhao, T. (2026). Assessing nighttime homeless shelter accessibility and its socioeconomic associations in Los Angeles. *Journal of Transport Geography*, *131*, 104553. (JCR Q1, IF=6.3) [link].
2. **Yi, S*.,** Tu, W., Zhao, T., Li, X., Zhang, Y., Li, D., ... & Sun, Y. (2025). Uncovering nonlinear urban factors of homelessness: Evidence from New York City using interpretable machine learning. *Environment and Planning B: Urban Analytics and City Science*, 23998083251342406. (JCR Q1, IF=2.6) [link].
3. **Yi, S*.,** Li, X., Li, D., Dong, X., Wang, R., & Xu, Q. (2025). Hyperlocal heat stress around bus stops in Philadelphia: Insights from spatio-temporal microclimate modeling and explainable AI. ***Computers, Environment and Urban Systems***, *122*, 102341. (JCR Q1, IF=8.3) [link].

4. **Yi, S*.,** Li, X., Tu, W., & Zhao, T. (2025). Planning for cooler cities: A multimodal AI framework for hyperlocal spatio-temporal urban heat stress prediction and mitigation. *Urban Forestry & Urban Greening*, 129101. (JCR Q1, IF=6.7) [link].
5. **Yi, S*.**, Li, X., Liu, Y., Dong, X., & Tu, W. (2025). A sub-meter resolution urban surface albedo dataset for 34 US cities based on deep learning. *Scientific Data*, *12*(1), 1-15. (JCR Q1, IF=5.8) [link].
6. **Yi, S*.**, Li, X., Ma, C., Wang, R., Zhou, Y., Xu, Q., & Zhao, T. (2025). Assessing the differential impact of vegetated and built-up areas on heat exposure environment: A case study of Los Angeles. ***Building and Environment*,** 112538. (JCR Q1, IF=7.1) [link].
7. **Yi, S.**, Li, X., Wang, R., Guo, Z., Dong, X., Liu, Y., & Xu, Q. (2024). Interpretable spatial machine learning insights into urban sanitation challenges: A case study of human feces distribution in San Francisco. ***Sustainable Cities and Society***, *113*, 105695. (JCR Q1, IF=11.7) [link].
8. Zhao, T., Liang, X., Biljecki, F., Tu, W., Cao, J., Li, X., & **Yi, S***. (2025). Quantifying seasonal bias in street view imagery for urban form assessment: A global analysis of 40 cities. ***Computers, Environment and Urban Systems*,** *120*, 102302. (JCR Q1, IF=7.1) [link].
9. Cai, Z., Chen, S., Ye, Y., Zhao, T., Yu, J., **Yi, S.**, Cao, J. (2026). Decoding urban soundscapes: spatial prediction and influence mechanism analysis with interpretable semi-supervised learning. ***Computers, Environment and Urban Systems, 126***, 102412. (JCR Q1, IF=8.3) [link].
10. Mueller, M., Quaye, I., **Yi, S.,** Foley, J., Shah, R., Li, X., ... & Hoque, S. (2026). Developing a machine learning model to map new-build gentrification: A mixed-methods approach. *PloS one*, *21*(1), e0341844 [link].
11. Zhu, L., Zhao, T., Cao, J., **Yi, S.,** Liu, S., Tu, W., & Zhang, H. (2025). Adaptive dynamic graph learning for forecasting urban multimodal flow. *International Journal of Geographical Information Science*, 1-26. (JCR Q1, IF=6.0) [link].
12. Wang, R., Sun, M. K., **Yi, S.,** Grekousis, G., & Dong, G. H. (2025). Exploring the associations between street-view green space quantity and quality, and influenza in Guangzhou, China through machine learning and spatial regression: A socio-economic equity perspective. ***Environment and Planning B: Urban Analytics and City Science,*** 23998083241312272**.** (JCR Q1, IF=2.6) [link].
13. Guo, X., Hu, Y., Zhang, Y., **Yi, S.,** & Tu, W. (2025). Evaluating the Quality of High-Frequency Pedestrian Commuting Streets: A Data-Driven Approach in Shenzhen. ***Smart Cities***, *8*(3), 83. (JCR Q1, IF=7.0) [link].
14. Dong, X., Ye, Y., Su, D., **Yi, S.,** Yang, R., Haase, D., & Lausch, A. (2025). Adaptive ranking of specific tree species for targeted green infrastructure intervention in response to urban hazards. ***Urban Forestry & Urban Greening***, 128776. (JCR Q1, IF=6.0) [link].
15. Ma, C., Zhang, Y., **Yi, S.,** & Lu, Y. (2025). Optimizing urban agricultural waste planning and management to enhance sustainability: Strategies for three types of cities. ***Sustainable Cities and Society***, 106168. (JCR Q1, IF=10.7) [link].
16. Yang, W., Xu, Q., ... **Yi, S**. Are different TOD circles oriented towards sustainability amidst urban shrinkage? Evidence from urban areas to suburbs in the Tokyo metropolitan area. ***Journal of Environmental Management***, *372*, 123274. (JCR Q1, IF=8.0) [link].
17. Yang, W., Xu, Q., **Yi, S.,** Shankar, R., & Chen, T. (2024). Enhancing transit-oriented development sustainability through the integrated node-place-ecology (NPE) model. ***Transportation Research Part D: Transport and Environment***, *136*, 104456. (JCR Q1, IF=7.3) [link].
18. Dong, X., Yang, R., Ye, Y., **Yi, S.**, Haase, D., & Lausch, A. (2024). Planning for green infrastructure by integrating multi-driver: Ranking priority based on accessibility equity. ***Sustainable Cities and Society***, *114*, 105767. (JCR Q1, IF=11.7) [link].

19. Chen, X., Tu, W., Yu, J., Cao, R., **Yi, S.**, & Li, Q. (2024). LCZ-based city-wide solar radiation potential analysis by coupling physical modeling, machine learning, and 3D buildings. ***Computers, Environment and Urban Systems***, *113*, 102176. (JCR Q1, IF=7.3) [link].
20. Cui, Q., Tan, L., Ma, H., Wei, X., **Yi, S.**, Zhao, D., ... & Lin, P. (2024). Effective or useless? Assessing the impact of park entrance addition policy on green space services from the 15-minute city perspective. ***Journal of Cleaner Production***, 142951. (JCR Q1, IF=11.1) [link].
21. Tu, W., Ye, H., Mai, K., Zhou, M., Jiang, J., Zhao, T., **Yi, S.,** & Li, Q. (2024). Deep online recommendations for connected E-taxis by coupling trajectory mining and reinforcement learning. ***International Journal of Geographical Information Science***, *38*(2), 216-242. (JCR Q1, IF=5.7) [link].

PRESENTATION

1. Presenter "Planning for cooler cities: A multimodal AI framework for hyperlocal spatio-temporal urban heat stress prediction and mitigation." The 2026 American Association of Geographers' Annual Meeting. In-Person. March 18, 2026.
2. Presenter "Driverless Cars and the City: Associations Between the Urban Environment and Robotaxi Crash Density in San Francisco." The 2026 Transportation Research Board Annual Meeting. In-Person. Jan 12, 2024.
3. Presenter "Riding Through the Heat: Exploring Spatio-Temporal Heat Stress and Bike-Share Usage Patterns in New York City." The 2026 Transportation Research Board Annual Meeting. In-Person. Jan 12, 2026.
4. Presenter "Spatio-Temporal Heat Stress at Bus Stops in Philadelphia: Microclimate Modeling and Explainable AI." The 2025 Association of Collegiate Schools of Planning. In-Person. October 25, 2025.
5. Presenter "Assessing the differential impact of suburban vegetated areas and urban built-up on heat exposure environment: A case study in Los Angeles." The 2024 American Association of Geographers' Annual Meeting. In-Person. April 16, 2024.

RESEARCH EXPERIENCE

Associate Research Fellow Dec 2024 –
Leonard Davis Institute of Health Economics (Penn LDI), University of Pennsylvania
- Leveraging AI and high-resolution spatio-temporal modeling to explore the impacts of heat exposure on vulnerable populations and public health outcomes.

Graduate Researcher June 2023 –
Advisor: Prof. Xiaojiang Li, Urban Spatial Informatics Lab, University of Pennsylvania
- Mapping and Modeling Heat Exposure at Hyperlocal Level (NSF #2314709).
- Aiming to combine Science, Data, and Design together to tackle the pressing urban challenges through collaboration with communities.

Advisor: Prof. Hamil Pearsall, Temple University Jan 2024 –
- Interactions of Sustainable Urban Design with Gentrification Processes (NSF #2312048, Co-PI).
- Collected millions geolocated Google Street View images to create before-and-after pairs for the same buildings.
- Developed a deep learning method to detect changes between the paired images and predict their categorization of gentrification.

Research Intern
Advisor: Prof. Jinhua Zhao & Prof. Shenhao Wang, MIT JTL Urban Mobility Lab June 2023 – Jan 2024

- Conducted research projects focusing on investigating the impacts of misallocation of bus drivers by looking at socioeconomic characteristics associated with bus routes.
- Measured and investigated relationship between service quality and service area socioeconomic indicators and characteristics.

WORKING EXPERIENCE

Data Scientist Intern May 2023 – Aug 2023
CityDNA Holdings Ltd Beijing, China

- Developed a universal simplification method for disorganized OSM road networks, producing a streamlined main road network suitable for non-precision applications like land division.
- Conducted network analysis using NetworkX library for urban infrastructure evaluation.
- Participated in an urban diagnostics project, utilizing Python to calculate various city indicators.

Machine Learning Researcher Intern Sept 2021 – Jun 2022
Advisor: Dr. Jixuan Cai, Tencent Holdings Ltd Shenzhen, China

- Integrated and analyzed multi-source spatio-temporal data through PySpark, including satellite images, open street maps, points of interest, and WeChat user trajectories.
- Helped to develop a multi-view neural network to detect fraudulent user behavior.

TEACHING EXPERIENCE

Teaching Assistant Jan 2025 –
University of Pennsylvania Philadelphia, US

- MUSA 6950 - AI for Urban Sustainability (Spring 2025).
- CPLN 5010 - Quantitative Planning Analysis Methods (Fall 2025).
- CPLN 6720 - Geospatial Data Science in Python (Fall 2025).
- CPLN 7900 - MUSA/Smart Cities Practicum (Spring 2026).

PROFESSIONAL SERVICE

Academic Journal Reviewer (more than 150 reviews)

- Nature Health
- Nature Cities
- Nature Communications
- npj Urban Sustainability
- Cell Reports Sustainability
- Computers, Environment, and Urban Systems
- Geophysical Research Letters
- Sustainable Cities and Society
- Cities
- Building and Environment
- International Journal of Applied Earth Observation and Geoinformation
- Applied Geography
- Journal of Transport Geography
- Travel Behaviour and Society
- And so on …

Mentoring

- Ruoyu Chen (UPenn undergraduate)
- Ziyi Guo (UPenn graduate student)

SKILLS

- **Language**: Chinese (native), English (fluent)
- **Programming Language**: Python, R, C++, JavaScript
- **Software**: Office Suites, Adobe Creative Suites, QGIS, ArcGIS, SPSS, ENVI